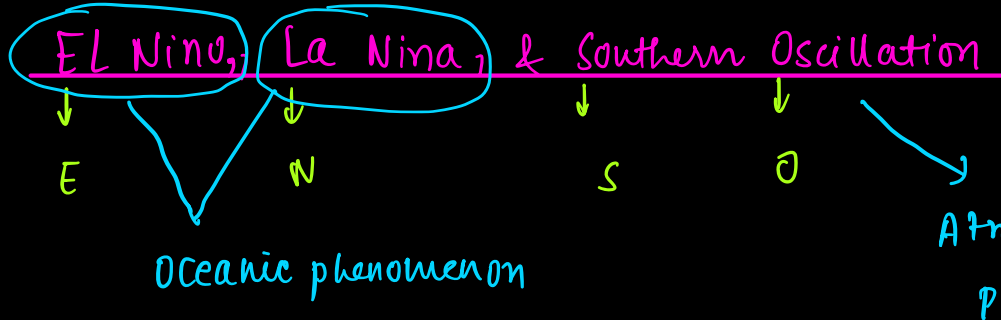




⇒ ENSO is a climate phenomena that occurs in the Pacific ocean.

Results of fluctuation of sea-surface temp. in equatorial western & eastern pacific.

El Nino → warmer than normal condⁿ. of Eastern pacific

La Nina → colder than normal condⁿ. of Eastern pacific

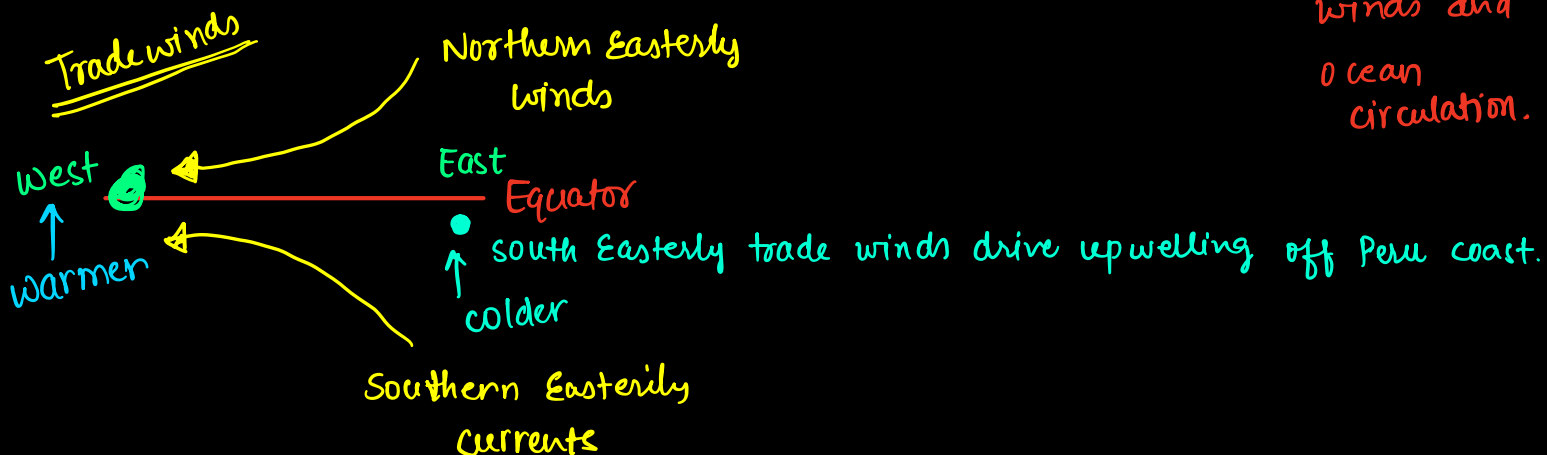
ENSO Cycle

1. Normal Cycle
2. La Nina
3. El Nino

$$\begin{aligned} T \uparrow &\Rightarrow P \downarrow \\ T \uparrow &\Rightarrow E \uparrow \end{aligned}$$

1) Normal condⁿ. in Equatorial Pacific —

- Western equatorial Pacific is warmer
 - Eastern equatorial Pacific is colder
- context of prevailing winds and ocean circulation.



- * Deep water carries nutrients for phytoplankton. Thriving phytoplankton attract zooplankton, fish & fisherman.

★ Mechanism of La Nina Process ———

- * Also referred as 'The Little Girl', 'anti-El Nino', or 'a cold event.'
- Generally La Nina follows an El Nino. It is characterised by unusually cold ocean temp. in the equatorial Pacific.
- * Increase in the intensity of Trade winds. Due to this, upwelling intensifies and temperature further drops down in the eastern Pacific, whereas, increased temp. is observed in the western Pacific. Therefore Walker Cells intensify.
 - * It is a process during which the Easterly trade winds becomes stronger than the normal driving more intense upwelling along the coast of Peru.

Walker Circulation or Walker Cell ———

- An atmospheric system of airflow in the equatorial Pacific ocean. caused by differences in heat distribution b/w ocean & land.
- * The Walker circulation is caused by the pressure gradient force that results from a high pressure system over the eastern Pacific ocean, and a low pressure system over Indonesia.
 - * The Walker circulations of tropical Indian, Pacific, and Atlantic basins results in westerly surface winds in northern summer.

in the first basin & easterly winds in the second & third basins.

⇒ As a result, the temp. structure of the three oceans display dramatic asymmetries.

→ The equatorial Pacific & Atlantic both have cool surface temp. in northern summer in the east, while cooler surface temperatures prevail only in the western Indian Ocean.

* These changes in surface temp. reflect changes in the depth of the thermocline.

→ During non-El Nino condⁿ, the Walker circulation is seen at the surface as easterly trade winds that move water and air warmed by the sun towards the west.

★ El Niño ——— (The Little Boy or the Christ child)

→ El Nino is a process during which the warm waters from the Western equatorial Pacific spread towards the Eastern equatorial Pacific.

* This warm water depresses the thermocline in the Eastern equatorial Pacific and thereby suppressing the upwelling.

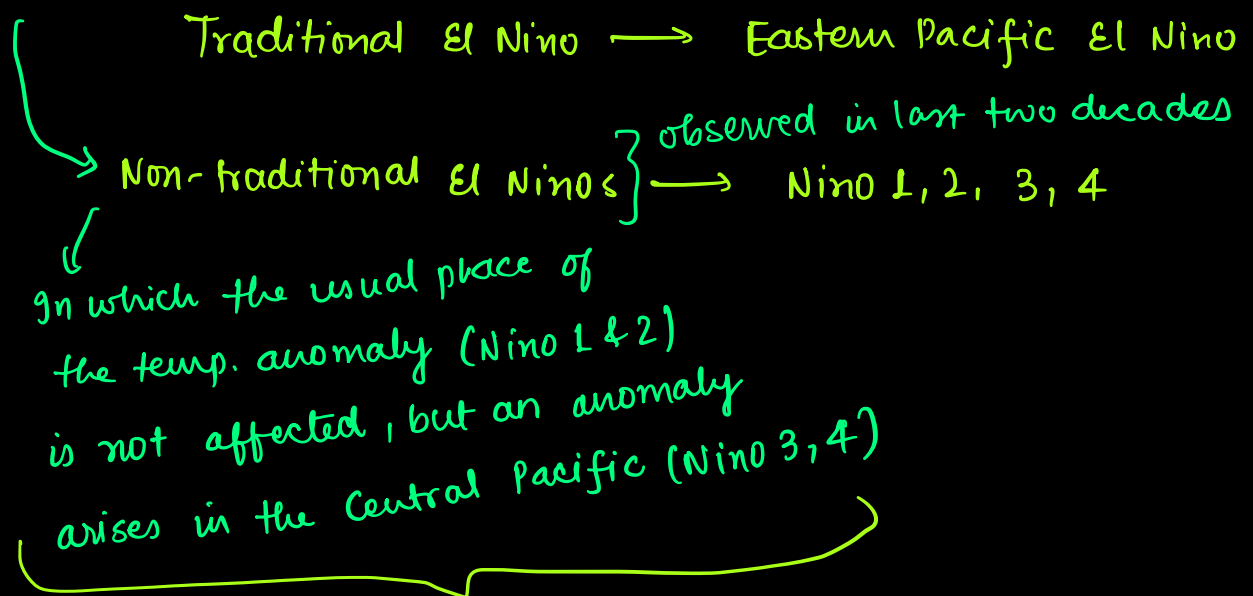
* During the El Nino convection shifts from western equatorial Pacific towards central equatorial Pacific.

→ It refers to the large scale ocean-atmosphere climate interaction linked to a periodic warming (around December) in sea surface temperatures across the central & east-central equatorial Pacific.

★ Mechanism —

→ Anti-walker Current during El Nino phase in the equatorial Pacific.

★ El-Nino Modoki —



phenomena is c/d Central Pacific El Nino dateline El Nino bcz the anomaly arises near the dateline.

★ Impacts of El Nino —

- ① Flood
- ② Draughts
- ③ Coral Bleaching
- ④ Poor Agriculture
- ⑤ Food Inflation
- ⑥ MSP issue (Budget Deficit)
- ⑦ Poverty (BPL)
- ⑧ Livelihood (Fishing industry)

★ Southern Oscillation ———

- * The southern oscillation describes a bimodal variation in sea level barometric pressure b/w observation stations at Darwin, Australia and Tahiti.
- It is quantified as Southern Oscillation Index (SOI), which is a standardized difference b/w the two barometric pressures.
- * The SO is the accompanying atmospheric component, coupled with the sea temp. change : El Nino is accompanied by high air surface pressure in the tropical western Pacific & La Nina with low air surface pressure there.
 - The two periods last several months each and typically occur every few years with varying intensity per period.

★ SOI ———

- * The SOI tracks the atmospheric part of the El Nino - La Nina climate pattern by comparing surface air pressure anomalies at Darwin, Australia, to pressure anomalies at Tahiti, French Polynesia.
- ⇒ The anomalies - departures from average conditions - indicate the strength of the Walker circulation.

Positive SOI —————→ La Nina

Negative SOI —————→ El Nino

known as
SO.

1. Rise in surface air pressure over Darwin (Australia)
2. Fall in air pressure over Tahiti.

Positive SOI

- ① Pressure at Tahiti is greater than that at Darwin
- ② High Pressure over eastern Pacific.
- ③ Drought condⁿ in the Eastern Pacific & good rainfall in western Pacific.
- ④ Good for Indian Monsoon.

Negative SOI

- ① Pressure at Darwin is greater than that at Tahiti.
- ② Low Pressure over eastern Pacific.
- ③ Drought condⁿ in the western Pacific & good rainfall in eastern Pacific
- ④ Bad for Indian Monsoon.

★ ENSO —

→ El Nino - Southern Oscillation (ENSO) is an irregular periodic variation in winds and sea surface temperatures over the tropical eastern Pacific Ocean, affecting the climate of much of the tropics and subtropics. The warming phase of the sea temp. is known as El Nino and the cooling phase as La Nina